CIVIL SYMPOSIUM

STUDY GUIDE

INGENIUM

ROUND 1 - EGG DROP CHALLENGE

In this challenge, teams will be tasked with designing and constructing a protective structure that allows an egg to survive a drop from a designated height. Teams will be provided with a fixed selection of materials and a limited amount of time to design, build, and test their structures.

The objective is not only to protect the egg, but to develop an efficient engineering solution while working within the given material and time constraints. Teams will be expected to consider factors such as impact force, energy absorption, structural stability, weight distribution, and material selection when designing their structures.

During the challenge, teams will be subjected to undisclosed distractions and additional conditions introduced by the organizers. These may require teams to adapt their designs or respond to unexpected changes during the challenge. The exact nature of these distractions will not be revealed beforehand, requiring teams to account for uncertainty and demonstrate adaptability.

Teams will be evaluated based on the survival of the egg, structural performance, efficient use of materials, quality of engineering design, and their ability to adapt to unexpected conditions. Additional points may be awarded for particularly innovative or well-engineered solutions.

The challenge is designed to **test engineering design, problem-solving, resource management, adaptability, and teamwork,** while encouraging teams to develop solutions that remain effective even when faced with unforeseen circumstances.

And a surprise at the end...

ROUND 2 - ENGINEERING PROBLEM SOLVING

In the Final Round, delegates will be presented with a real-world engineering problem at the beginning of the round. Each team will be given a fixed period of time to research the problem, analyze the available information, and develop a practical and innovative solution.

During the preparation period, teams will be expected to conduct research using reliable sources, identify the key challenges, analyze relevant data, perform necessary calculations, and support their proposed solution with sound engineering principles. Teams may use appropriate research tools and resources permitted by the organizers.

Following the preparation period, each team will present its findings and proposed solution to the judges and participating teams. Presentations should clearly explain the problem, research conducted, relevant data, engineering reasoning, proposed solution, advantages, limitations, and potential risks.

Teams will be evaluated on their understanding of the problem, quality and reliability of research, engineering analysis, use of data and calculations, practicality and creativity of their solution, clarity of presentation, and ability to defend their proposal under questioning.

The round is designed to simulate real-world engineering decision-making, where problems rarely have a single perfect answer. The strongest team will be the one that develops a well-researched, technically sound, practical solution and demonstrates the ability to justify and defend its decisions using evidence and logical reasoning.